Southeast University

Department of Computer Science and Engineering

LAB Report

Course Name: Computer Networking LAB

Course Code: CSE 342.2

Experiment No.: 02

Experiment Name: Basic Router Interface with SSH Configuration

Date of performance:

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Submitted By,

Name: MD. Kamran Hasan Ratin

ID: 2022000000066

Section: 02

Batch:

Experiment No: 02

Experiment Name: Basic Router Interface configuration and Secure Remote Access using SSH Implementation.

Objective: The Primary focus of this experiment is to secure a Router against unauthorized access by configuring a localized, encrypted authentication database and applying strict SSH Transport Protocol for remote access management. These critical security commands are executed out-of-band via a direct console connection from a end-device. Secondary objective is to include basic IPV4 interface configuration and connectivity verification with other end-devices.

Network Topology:

- 1x Cisco 2911 Router
- 2x Laptop for interface
- 1x Switch
- 3x PC
- Console Cable
- Straight-through Cable

IP Addressing:

- Routerr : Network 192.168.0.0/25
- Default Gateway 192.168.0.1/25

CLI Command on Routerr's Interface:

For this we need to open Desktop Interface on Laptop and Then open terminal Interface for write Commands.

```
Routen > enable
Routen# Configune Terminal
Routen(config)# username RAFIN Password 111I
Routen (config)# Do write memory
Routen (config)# line console 0
Routen (config)# login local
Routen (config)# exec-timeout 2
Routen (condig)# service Password-encryption
Routen (config)# exit
```

To Assign IP Address:

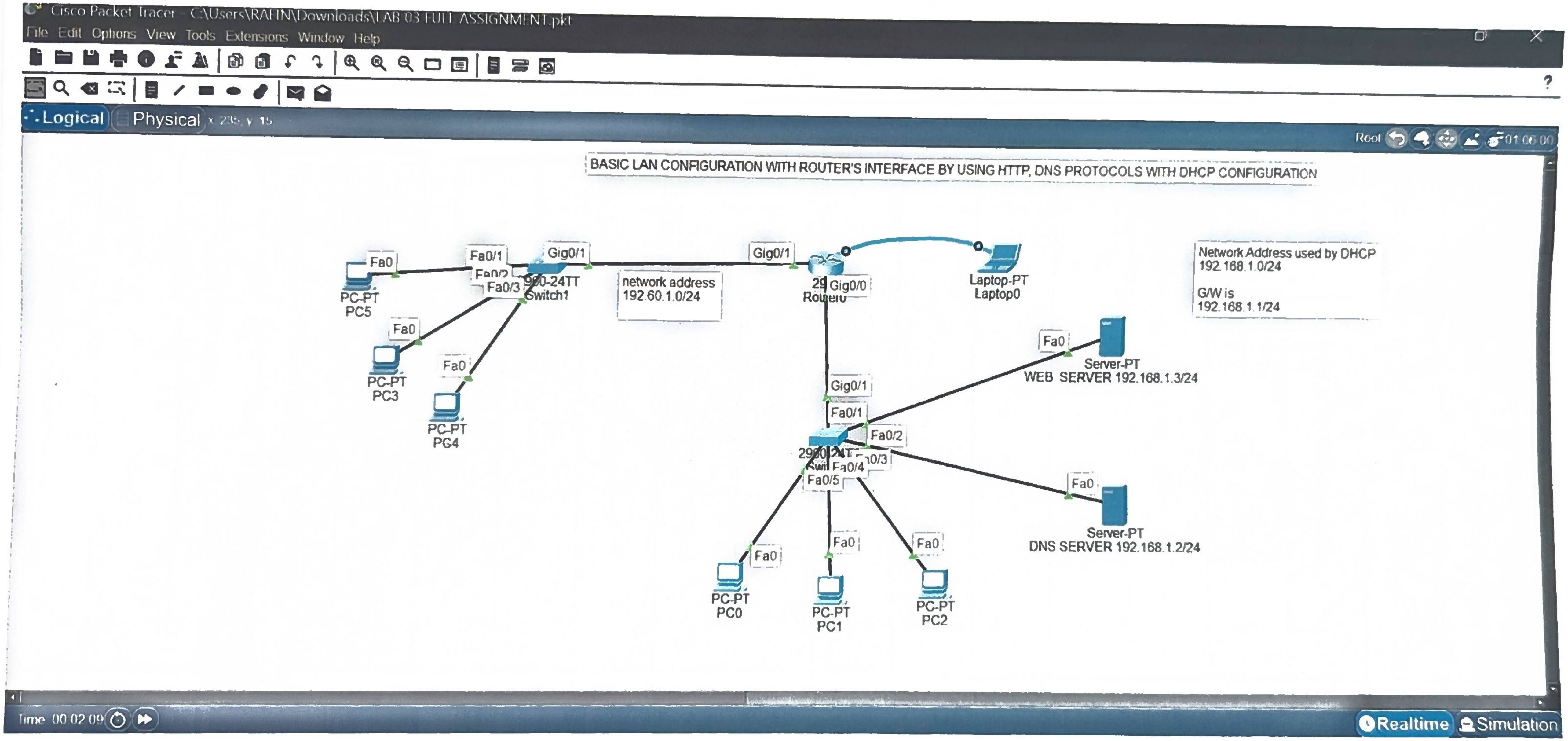
```
Routen(config)# Interface gig 0/0
Routen(config)# IP address 192.168.0.1 255.255.255.128
Routen (config)# no shutdown
Routen (config)# exit
```

* We also need to assign IP address in end-devices in static way.

4 Commands for SSH configuration: open Laptop on Router's Interface Laptop:

```
Router> enable
Router# configure Terminal
Router(config)# hostname R1
R1(config)# IP domain-name net-lab.local
R1 (config)# crypto key generate rsa
R1 (config)# username RAFINSEU privilege 15 secret 1234
R1 (config)# line vty 0 15
R1 (config)# login local
R1 (config)# Transport input ssh
R1 (config)# end
R1 (config)# Do write memory
R1 (config)# exit.
```

* We need to turn on ports on switch as well. AND also need to give same SSH Commands on switch to complete SSH configuration. switch is a layer 2 device so it does not need an IP address so we just need to configure basic SSH Protocol


Cisco Packet Tracer - C:\Users\RAFIN\Downloads\LAB 03 FULL ASSIGNMENT.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
Root
BASIC LAN CONFIGURATION WITH ROUTER'S INTERFACE BY USING HTTP, DNS PROTOCOLS WITH DHCP CONFIGURATION
Fa0
PC-PT
PC5
Fa0/1
Fa0/3
Gig0/1
Switch1
Fa0
PC-PT
PC3
Fa0
PC-PT
PC4
network address
192.60.1.0/24
Gig0/1
Gig0/0
Router0
Laptop-PT
Laptop0
Network Address used by DHCP
192.168.1.0/24
G/W is
192.168.1.1/24
Fa0
Server-PT
WEB SERVER 192.168.1.3/24
Gig0/1
Fa0/1
Fa0/2
Fa0/4
Fa0/5
Fa0
Server-PT
DNS SERVER 192.168.1.2/24
Fa0
PC-PT
PC0
Fa0
PC-PT
PC1
Fa0
PC-PT
PC2
Time: 00:02:09
Realtime
Simulation

To verify Connectivity and Remote Access:

- For Connectivity: Go to Command Prompt Interface on end-device

  - $\boxed{\text{Ping 192.168.0.1}}$ → You should receive four successfull replies.

- Remote Access Test:

  - ssh -l username IP-address

→ ssh -l RAFINSEU 192.168.0.1 → This will enable the Remote Access.

Conclusion:

This experiment is successfully performed Basic Router LAN configuration and Remote Access SSH Protocol. After giving the needed Commands we tests and verify the connectivity and remote access both are working perfectly in Cisco Packet tracer.